

An Interval-Certified Positivity Theorem for a Corrected GLSM Metric along a Calabi–Yau Continuation Path

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Abstract

We give a computer-assisted proof of positive definiteness for a frozen corrected gauged-linear-sigma-model (GLSM) quotient metric along a connected one-parameter path on an explicit compact toric Calabi–Yau threefold with $(h^{2,1}, h^{1,1}) = (5, 81)$. A parameterized Krawczyk construction proves existence and local uniqueness of the implicit hypersurface–moment-map branch for $d \in [1.49985, 1.5]$. The branch is covered by seven root patches and seven uniqueness bridges, with every componentwise Krawczyk ratio below 0.85. Positive definiteness of the corrected 3×3 Hermitian pullback metric is then established on 1003 exactly adjacent interval segments by rational lower bounds for its three leading principal minors. A separate formula implementation reconstructs the weakest third-minor segment in each of six metric patches, and an independent 83-real endpoint construction recomputes the corrected determinants using exact rational interval arithmetic. At the endpoint a fixed direction ensemble also proves a relative correction larger than 1.83%. The result is a theorem about a specified frozen $\partial\bar{\partial}$ ansatz on a one-dimensional path. It is not a global Ricci-flat metric, a derivation of the physical α' correction, or a complete string vacuum.

1 Introduction

Yau’s theorem guarantees a Ricci-flat Kähler metric in each Kähler class of a compact Calabi–Yau manifold, but explicit metrics on compact threefolds are generally unavailable. Numerical work based on balanced metrics and energy functionals made these metrics computationally accessible [1, 2, 3, 4]. Machine-learning methods have subsequently enlarged the range of geometries and moduli that can be studied [5, 6, 7]. These methods measure approximation quality by sampled or integrated residuals; they do not ordinarily produce mathematically rigorous enclosures at every point of a parameter interval.

Corrections to an internal Calabi–Yau metric are directly relevant to string compactifications. In particular, Ref. [8] computes a leading α'^3 correction to a numerically approximated quintic metric. That work is the closest physical antecedent to the present calculation, but its neural approximation and percent-level residual are not interval certificates. Rigorous interval methods have entered nearby Calabi–Yau problems, for example the computer-assisted monodromy calculation for a toric K3 family in Ref. [9], but we found no previous interval proof of corrected-metric positivity on an explicit compact Calabi–Yau continuation branch. To the best of our knowledge, the result below is the first computer-assisted interval-arithmetic positivity theorem for a nontrivially corrected GLSM quotient metric along a connected continuation path on an explicit compact Calabi–Yau threefold used in a string-compactification

model. This priority statement concerns the combination, not any individual GLSM, interval, or Krawczyk ingredient.

The threefold is the $(5, 81)$ example occurring in the explicit type-IIB compactifications of Ref. [10]. The role of this provenance is to make the geometry non-toy and fully identifiable. It does not imply that the local metric theorem below certifies the full vacuum constructed in that reference.

The central distinction of this paper is between discovery and proof. The correction coefficients were selected by a finite-sample approximation experiment and then frozen as exact binary rationals. Their origin does not prove that the resulting tensor is positive. Positivity follows only from a subsequent interval calculation on a Krawczyk-certified implicit branch.

2 Geometry and frozen metric ansatz

Let $Q \in \mathbb{Z}^{81 \times 85}$ be the disclosed GLSM charge matrix and let $r \in \mathbb{R}^{81}$ be the FI vector. In a regular toric chart, write z_i for the Cox coordinates and solve the moment-map equations

$$QD = r, \quad D_i = |z_i|^2 \exp((Q^T u)_i), \quad i = 1, \dots, 85, \quad (1)$$

for $u \in \mathbb{R}^{81}$. With $D = \text{diag}(D_i)$, the horizontal ambient quotient tensor is

$$H = D - DQ^T(QDQ^T)^{-1}QD. \quad (2)$$

The identities $Q[:, \text{divisor basis}] = I_{81}$ and $Qv_i = 0$ fix the linear identification with the toric Kähler data. If A is the analytic tangent matrix obtained from the implicit hypersurface equation, the baseline pullback metric can equivalently be evaluated as

$$G_0 = A^*DA - (QDA)^*(QDQ^T)^{-1}(QDA). \quad (3)$$

The correction is defined by a scalar function on the quotient. Let

$$S = \sum_{i=1}^{85} D_i, \quad p_i = D_i/S, \quad (4)$$

and let $S_0, p_i^{(0)}$ denote their values at the disclosed reference point. For the archived rational direction $v \in \mathbb{Q}^{85}$ define

$$q = \frac{S - S_0}{S + S_0}, \quad s = \sum_{i=1}^{85} v_i (p_i - p_i^{(0)})^2. \quad (5)$$

The frozen correction potential is

$$\phi = c_1(\lambda_1 q^2)^2 + c_2(\lambda_2 s)^2, \quad G_{\text{corr}} = G_0 + \partial\bar{\partial}\phi, \quad (6)$$

where $c_1, c_2, \lambda_1, \lambda_2, v, S_0$, and all $p_i^{(0)}$ are serialized as reduced rational pairs. In particular,

$$c_1 = -\frac{3704684168964555}{9007199254740992}, \quad c_2 = \frac{4163219474253507}{1125899906842624}. \quad (7)$$

Because Eq. (6) is a $\partial\bar{\partial}$ deformation of a scalar that is invariant under the complexified GLSM action, it preserves the local Kähler structure wherever the implicit quotient construction is regular. The theorem below addresses positive definiteness, not the size of a global Ricci or Monge–Ampère residual.

3 Certified implicit branch

The parameter d describes the disclosed affine continuation, and we set $t = d - 3/2$. At each parameter patch the coupled implicit system consists of the real and imaginary hypersurface equations and the 81 moment-map equations, hence 83 real unknowns. Denote this system by

$$F(t, x) = 0, \quad x \in \mathbb{R}^{83}. \quad (8)$$

For a parameter interval T , center x_0 , box $X = x_0 + [-r_1, r_1] \times \cdots \times [-r_{83}, r_{83}]$, and a midpoint inverse C , the parameterized Krawczyk image is

$$K(T, X) = x_0 - CF(T, x_0) + (I - CF_x(T, X))(X - x_0). \quad (9)$$

Strict componentwise inclusion $K(T, X) \subset \text{int } X$ proves existence and uniqueness in the declared box for each $t \in T$; overlap bridges identify neighboring local branches [11, 12].

The archived chain uses seven patches and seven bridges to cover

$$t \in [-3/20000, 0]. \quad (10)$$

The maximum exact-rational replayed image-to-radius ratios are

$$q_{\text{patch}} < 0.849476916, \quad q_{\text{bridge}} < 0.829319820, \quad (11)$$

and both are strictly below the preregistered threshold $17/20$. Every patch and bridge contains 83 componentwise strict inequalities. Overlap and an independent endpoint enclosure identify one base-anchored branch throughout the interval.

Proposition 1 (Certified continuation branch). *For every $d \in [1.49985, 1.5]$ the disclosed 83-real implicit system has a solution in the corresponding archived Krawczyk box. The solution is unique inside that box, and the overlap certificates identify these local solutions as one connected branch.*

The proposition does not assert that no other branch exists outside the archived boxes.

4 Metric interval theorem

On each subinterval, interval implicit differentiation of the hypersurface and moment-map systems gives all six real first and second derivatives needed for the 3×3 Hermitian pullback tensor. The inverse of QDQ^T and the joint implicit Jacobian is enclosed through a scaled finite Neumann series with a rigorous geometric tail. The metric is symmetrized by intersection with its Hermitian transpose, and Eq. (6) is differentiated analytically with second-order interval jets.

Let $\Delta_k(d)$ be the determinant of the leading $k \times k$ principal submatrix of $G_{\text{corr}}(d)$. By the Hermitian Sylvester criterion,

$$G_{\text{corr}}(d) \succ 0 \iff \Delta_1(d) > 0, \Delta_2(d) > 0, \Delta_3(d) > 0. \quad (12)$$

The metric layer consists of six patches and 1003 exactly adjacent segments. The portable verifier recomputes the connected union and the exact minimum serialized lower bounds displayed in Table 1.

Theorem 1 (Corrected-metric positivity on the continuation path). *Along the base-anchored branch of Proposition 1, the frozen corrected pullback metric of Eq. (6) is positive definite for every*

$$d \in [1.49985, 1.5]. \quad (13)$$

Certified quantity	Result
Depth interval	[29997/20000, 3/2]
Metric patches / segments	6/1003
Root patches / bridges	7/7
Minimum lower bound for Δ_1	7012202881170207/576460752303423488
Minimum lower bound for Δ_2	3516194174689203/9223372036854775808
Minimum lower bound for Δ_3	7217687928450223/302231454903657293676544
Maximum root-patch ratio	4808980629275713/5661108081689003
Maximum root-bridge ratio	5163879392018433/6226644134068241

Table 1: Exact rational outputs of the portable certificate replay.

Proof. The six metric covers contain 1003 nondegenerate rational intervals. Exact end-point comparison shows adjacency within each patch, overlap between patches, and union [29997/20000, 3/2]. On every segment, interval evaluation gives strict positive rational lower bounds for Δ_1 , Δ_2 , and Δ_3 . Their minima are the positive fractions in Table 1. Proposition 1 identifies the implicit solution boxes with one branch. Equation (12) then proves positive definiteness at every parameter value on that branch. $\square$

5 Independent checks and correction materiality

Serialized replay can detect gaps, sign failures, and stale attachments, but it does not by itself detect a common formula error in the generator. We therefore add two checks with deliberately different boundaries.

First, a separate Arb implementation imports neither the primary metric generator nor its adapter. Before reconstruction, the weakest serialized Δ_3 segment is selected in each of the six metric patches. The second implementation rebuilds Eq. (3), the correction jets, and all three principal minors on these six segments. All 18 independent lower bounds are positive and contain the corresponding primary lower bounds. These checks reuse the primary certified root boxes and are therefore a formula replication, not a second proof of the entire continuation chain.

Second, at $d = 1.5$ a separate 83-real Krawczyk box is constructed. Its 83 strict inclusions prove a locally unique root, and the continuation endpoint box lies componentwise inside this uniqueness box with maximum radius ratio 10^{-4} . Independent baseline and correction implementations serialize entrywise directed rational bounds. A stdlib-only checker then adds the complex interval matrices and recomputes all three determinants by the Leibniz formula using exact `fractions.Fraction` arithmetic. The resulting lower bounds are positive.

To exclude a merely formal correction whose certified effect is indistinguishable from zero, a fixed ensemble of 31 Gaussian-integer directions is evaluated at the endpoint. For two directions the rigorous lower bound on

$$\frac{|v^*(G_{\text{corr}} - G_0)v|}{v^*G_0v} \quad (14)$$

exceeds one percent. The largest certified lower bound is

$$\frac{7568816278373741}{413272176438128128} = 0.0183143620\dots \quad (15)$$

This is an existence statement within a fixed finite direction ensemble, not an operator-norm estimate along the full path.

6 Reproducibility and proof architecture

The release separates the heavy generator layer from the portable review layer. The heavy calculations use Arb midpoint-radius arithmetic [13], analytic implicit derivatives, and exact rational serialization. The public archive contains the resulting directed bounds, the six metric covers, root bridges, endpoint witnesses, independent formula checks, and their complete transitive SHA-256 provenance graph. The graph contains 88 visited nodes and 226 resolved edges, with no unresolved or ambiguous proof edge. Three failed diagnostics are retained in a historical manifest but excluded from proof traversal.

The command

```
python verify_release.py
```

requires only Python 3.11 or later. It verifies the release manifest, checks the exact rational continuation and positivity inequalities, independently recomputes the endpoint determinants and materiality ratios, and runs mutation tests. The mutations delete a metric segment, replace a positive minor by zero, and force equality in a bridge inclusion; every mutation must be rejected.

The portable checker is intentionally not described as formal verification. It trusts the correctness of Python integer arithmetic, SHA-256, the serialized Arb enclosures, and the stated analytic formulas. Conversely, the heavy generator is not needed for last-mile replay and cannot silently change after publication because every included byte is bound by the release manifest.

7 Scope, interpretation, and limitations

The strongest admitted conclusion is Theorem 1. Several nearby statements are not consequences of it.

First, the parameter interval is one-dimensional and has length 1.5×10^{-4} . Each metric evaluation reconstructs all transverse derivatives needed for a three-complex-dimensional pullback metric, but this does not prove positivity on an open six-real-dimensional tube. Nor does it cover the entire compact hypersurface or all toric charts.

Second, G_0 is a section-free GLSM quotient reference metric, not a certified Ricci-flat Calabi–Yau metric. The correction of Eq. (6) is a frozen gauge-invariant approximation ansatz. The theorem proves that this specified correction preserves positive definiteness on the declared path. It does not derive the ansatz from the ten-dimensional effective action, bound omitted α' terms, or certify a global Ricci/Monge–Ampère error.

Third, the phrase “the (5,81) model” identifies an explicit geometry from a realistic compactification program. The result does not certify mobile D3 fields, Pfaffians, uplift, a 113-field mass matrix, or phenomenology. Accordingly, neither “full physical vacuum” nor “global corrected Calabi–Yau metric” is an admissible description of this theorem.

8 Conclusion

We have exhibited a reviewer-replayable computer-assisted theorem in a regime where numerical Calabi–Yau work has usually relied on sampled accuracy. A Krawczyk-certified 83-real branch, a gap-free 1003-segment metric cover, strict Sylvester bounds, stratified formula replication, and an independent endpoint determinant calculation together prove that a nonzero frozen correction preserves metric positivity throughout a connected path on the explicit (5,81) geometry. The natural next mathematical extension is an open tubular neighborhood and then multi-chart coverage. Those extensions are separate results and are not assumed here.

Data and code availability

The versioned release contains the manuscript, exact proof records, portable checker, mutation tests, trust model, and SHA-256 manifest. Code is released under the BSD 3-Clause License. Release v1.0.1 is available at the public GitHub repository and is permanently archived in the Zenodo record doi:10.5281/zenodo.21923926.

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